

COMPUTER NETWORKS

(25MC410)

I M.C.A II Semester

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The Medium Access Control Sublayer

- Many link-layer communications protocols rely on a broadcast communication medium to transmit data.
- Any such protocol requires additional mechanisms to allow multiple senders to efficiently and fairly share the broadcast medium.
- In any broadcast network, the key issue involves determining who gets to use the channel when there is competition for it.
- Broadcast channels are sometimes referred to as multiaccess channels or random access channels.
- The protocols used to determine who goes next on a multiaccess channel belong to a sublayer of the data link layer called the MAC (Medium Access Control) sublayer.

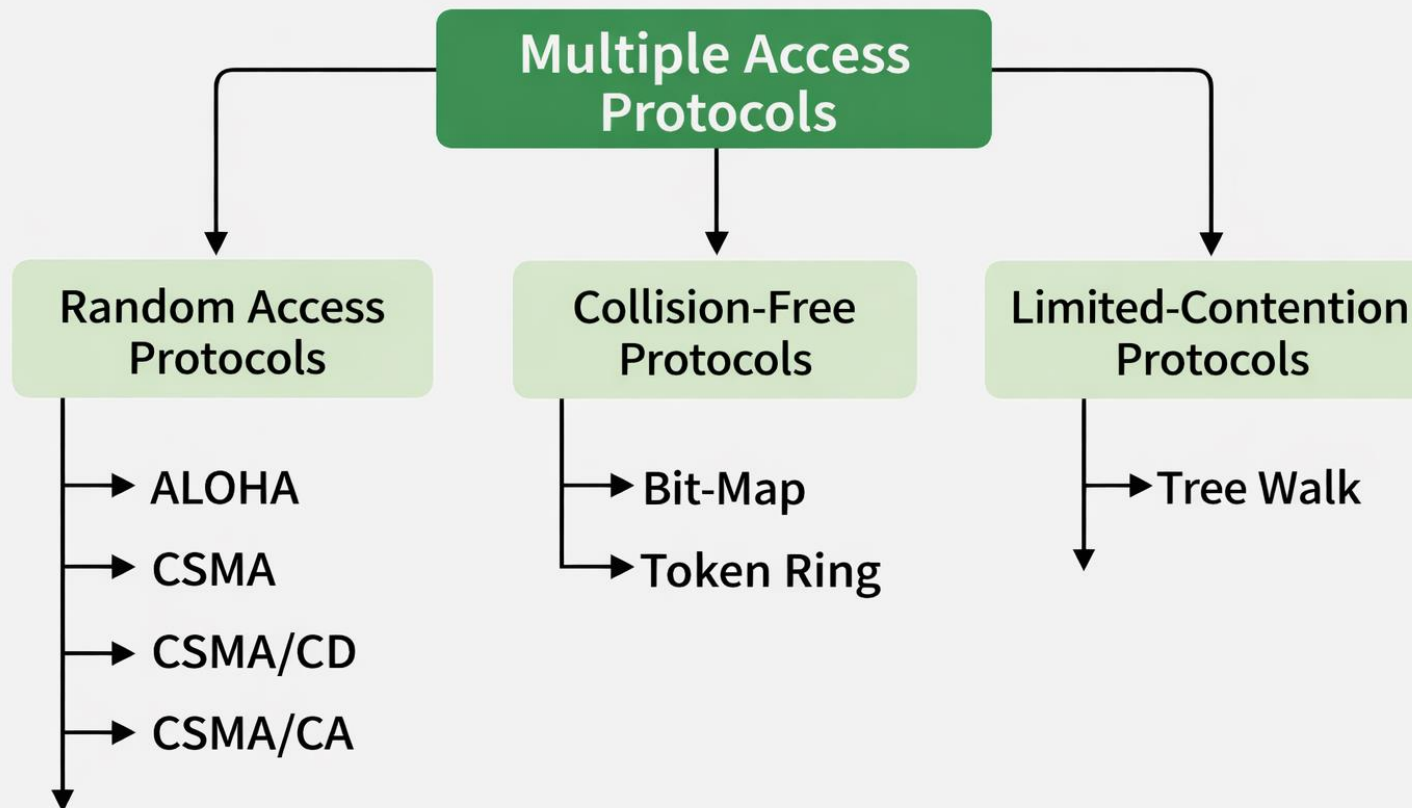
- The conventional way of allocating a single channel, such as a telephone trunk, among multiple competing users is to chop up its capacity by using one of the multiplexing schemes such as FDM (Frequency Division Multiplexing) or TDM (Time Division Multiplexing).
- If there are N users, the bandwidth is divided into N equal-sized portions, with each user being assigned one portion.
- Since each user has a private frequency band, there is now no interference among users.
- When there is only a small and constant number of users, each of which has a steady stream or a heavy load of traffic, this division is a simple and efficient allocation mechanism.
- When the number of senders is large and varying or the traffic is bursty, FDM presents some problems.

Assumptions for Dynamic Channel Allocation

- **Independent Traffic:** The model consists of N independent stations each with a program or user that generates frames for transmission. Once a frame has been generated, the station is blocked and does nothing until the frame has been successfully transmitted.
- **Single Channel:** A single channel is available for all communication. All stations can transmit on it and all can receive from it.
- **Observable Collisions:** If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a collision. All stations can detect that a collision has occurred. A collided frame must be transmitted again later. No errors other than those generated by collisions occur.

Assumptions for Dynamic Channel Allocation

- **Continuous or Slotted Time:** Time may be assumed continuous, in which case frame transmission can begin at any instant. Alternatively, time may be slotted or divided into discrete intervals.
- **Carrier Sense or No Carrier Sense:** With the carrier sense assumption, stations can tell if the channel is in use before trying to use it. No station will attempt to use the channel while it is sensed as busy. If there is no carrier sense, stations cannot sense the channel before trying to use it. They just go ahead and transmit. Only later can they determine whether the transmission was successful.



Multiple Access Protocols:

- ALOHA: The story of our first MAC protocol starts out in pristine Hawaii, “pristine” can be interpreted as “not having a working telephone system.” Norman Abramson and his colleagues at the University of Hawaii who were trying to connect users on remote islands to the main computer in Honolulu.
- The one they found used short-range radios, with each user terminal sharing the same upstream frequency to send frames to the central computer.
- Their work has been extended by many researchers since then (Schwartz and Abramson, 2009).
- Although Abramson’s work, called the ALOHA system, used ground based radio broadcasting, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel.

Pure ALOHA:

- The basic idea of an ALOHA system is let users transmit whenever they have data to be sent.
- There will be collisions, of course, and the colliding frames will be damaged.
- If the frame was destroyed, the sender just waits a random amount of time and sends it again.
- The waiting time must be random or the same frames will collide over and over, in lockstep.
- Systems in which multiple users share a common channel in a way that can lead to conflicts are known as contention systems.

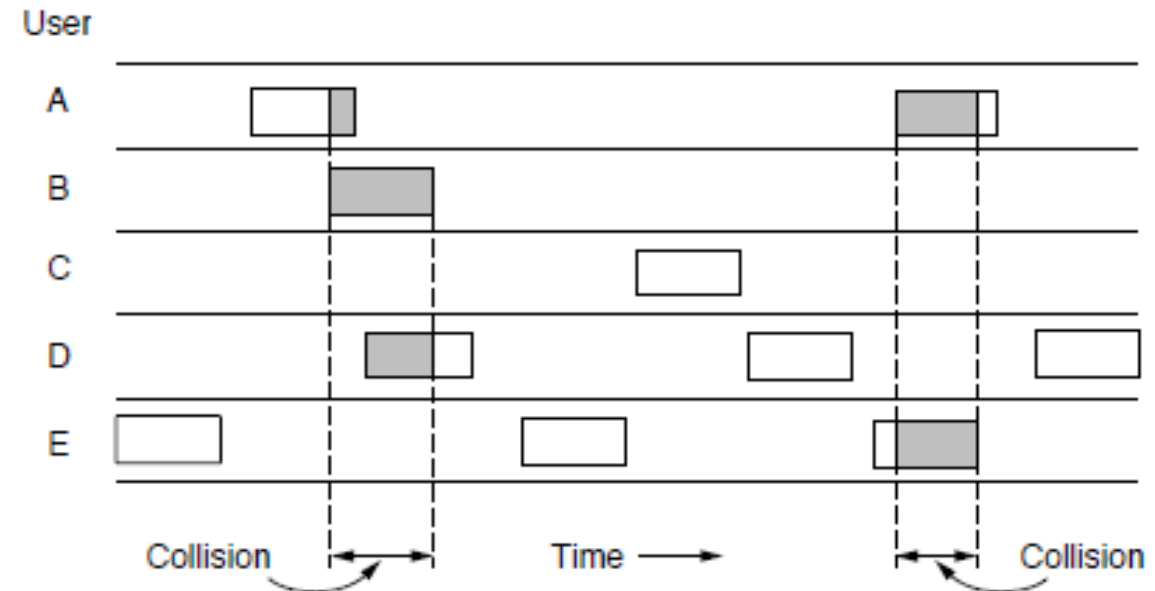


Figure 4-1. In pure ALOHA, frames are transmitted at completely arbitrary times.

Pure ALOHA:

- All the frames are of same length because the throughput of ALOHA systems is maximized by having a uniform frame size rather than by allowing variable-length frames.
- Whenever two frames try to occupy the channel at the same time, there will be a collision and both will be garbled.
- If the first bit of a new frame overlaps with just the last bit of a frame that has almost finished, both frames will be totally destroyed (i.e., have incorrect checksums) and both will have to be retransmitted later.

Pure ALOHA:

- Let the “frame time” denote the amount of time needed to transmit the standard, fixed-length frame.
- We assume that the new frames generated by the stations are well modeled by a Poisson distribution with a mean of N frames per frame time.
- If $N > 1$, the user community is generating frames at a higher rate than the channel can handle, and nearly every frame will suffer a collision.
- For reasonable throughput, we would expect $0 < N < 1$.

Pure ALOHA:

- In addition to the new frames, the stations also generate retransmissions of frames that previously suffered collisions.
- Let us further assume that the old and new frames combined are well modeled by a Poisson distribution, with mean of G frames per frame time. Clearly, $G \geq N$.
- At low load (i.e., $N \approx 0$), there will be few collisions, hence few retransmissions, so $G \approx N$.
- At high load, there will be many collisions, so $G > N$.
- Under all loads, the throughput, S , is just the offered load, G , times the probability, P_0 , of a transmission succeeding—that is,

$S = GP_0$, where P_0 is the probability that a frame does not suffer a collision.

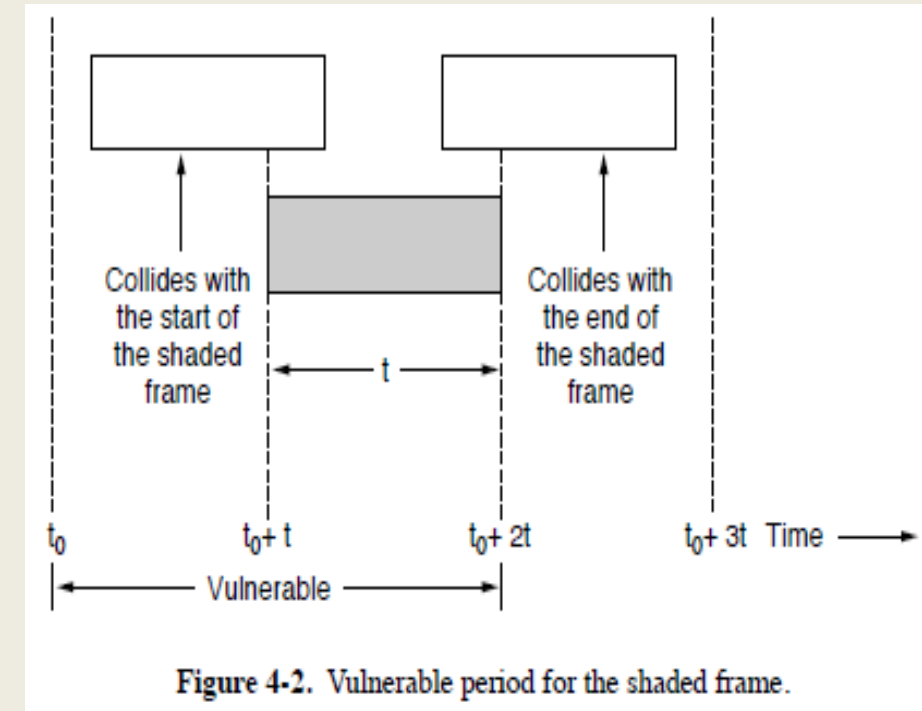
Pure ALOHA:

- The probability that k frames are generated during a given frame time, in which G frames are expected, is given by the Poisson distribution.

$$\Pr[k] = \frac{G^k e^{-G}}{k!}$$

- The probability of zero frames is just e^{-G} .
- In an interval two frame times long, the mean number of frames generated is $2G$.
- The probability of no frames being initiated during the entire vulnerable period is thus given by $P_0 = e^{-2G}$. Using $S = GP_0$, we get

$$S = Ge^{-2G}$$



Slotted ALOHA:

- Roberts (1972) published a method for doubling the capacity of an ALOHA system.
- His proposal was to divide time into discrete intervals called **slots**, each interval corresponding to one frame.
- This approach requires the users to agree on slot boundaries.
- In **slotted ALOHA** a station is not permitted to send whenever the user types a line.
- Instead, it is required to wait for the beginning of the next slot. Thus, the continuous time ALOHA is turned into a discrete time one.
- This halves the vulnerable period.
- The probability of no other traffic during the same slot as our test frame is then e^{-G} , which leads to

$$S = Ge^{-G}$$

Carrier Sense Multiple Access Protocols:

- Protocols in which stations listen for a carrier (i.e., a transmission) and act accordingly are called carrier sense protocols.

1- Persistent CSMA

Nonpersistent CSMA

P-Persistent CSMA

1- Persistent CSMA:

- When a station has data to send, it first listens to the channel to see if anyone else is transmitting at that moment.
- If the channel is idle, the stations sends its data.
- Otherwise, if the channel is busy, the station just waits until it becomes idle. Then, the station transmits a frame.
- If a collision occurs, the station waits a random amount of time and starts all over again. The protocol is called 1-persistent because the station transmits with a probability of 1 when it finds the channel idle.
- The propagation delay has a very important effect on collisions.
- There is a chance that just after a station begins sending, another station will become ready to send and sense the channel.
- If the first station's signal has not yet reached the second one, the latter will sense an idle channel and will also begin sending, resulting in a collision.

Nonpersistent CSMA:

- In nonpersistent CSMA a station senses the channel when it wants to send a frame, and if no one else is sending, the station begins doing so itself immediately.
- If the channel is already in use, the station does not continually sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission.
- Instead, it waits a random period of time and then repeats the algorithm.
- This algorithm leads to fewer collisions and better channel utilization but longer delays than 1-persistent CSMA.

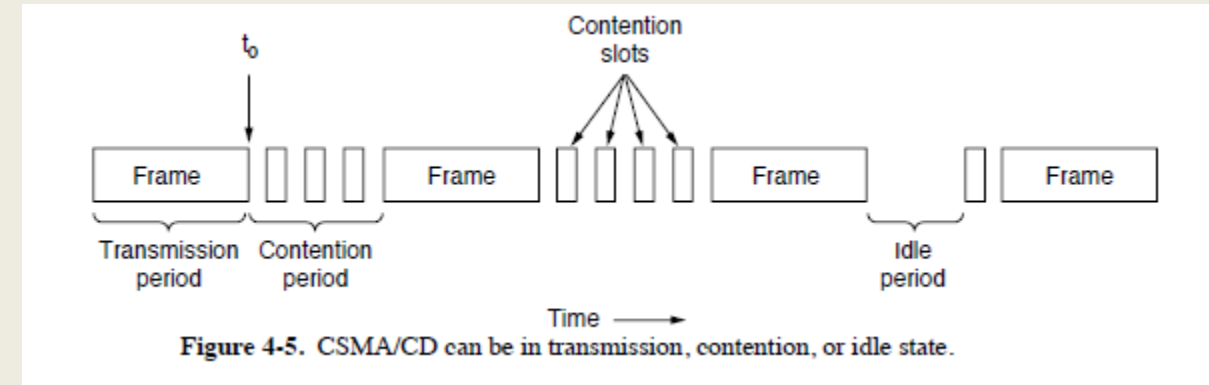
P-Persistent CSMA:

- It applies to slotted channels and works as follows. When a station becomes ready to send, it senses the channel.
- If it is idle, it transmits with a probability p .
- With a probability $q = 1 - p$, it defers until the next slot.
- If that slot is also Idle, it either transmits or defers again, with probabilities p and q . This process is repeated until either the frame has been transmitted or another station has begun transmitting.

Carrier Sense Multiple Access Protocols:

CSMA with Collision Detection:

- If two stations sense the channel to be idle and begin transmitting simultaneously, their signals will still collide.
- Another improvement is for the stations to quickly detect the collision and abruptly stop transmitting, (rather than finishing them) since they are irretrievably garbled anyway.
- This strategy saves time and bandwidth.



Collision- Free Protocols:

- We will examine some protocols that resolve the contention for the channel without any collisions at all, not even during the contention period.
- We assume that there are exactly N stations, each programmed with a unique address from 0 to $N - 1$.
- It does not matter that some stations may be inactive part of the time.
- We also assume that propagation delay is negligible.

A Bit-Map Protocol

Token Passing

Binary Countdown

A Bit-Map Protocol:

- The basic bit-map method, each contention period consists of exactly N slots.
- If station 0 has a frame to send, it transmits a 1 bit during the slot 0.
- No other station is allowed to transmit during this slot.
- Regardless of what station 0 does, station 1 gets the opportunity to transmit a 1 bit during slot 1, but only if it has a frame queued.
- After all N slots have passed by, each station has complete knowledge of which stations wish to transmit.

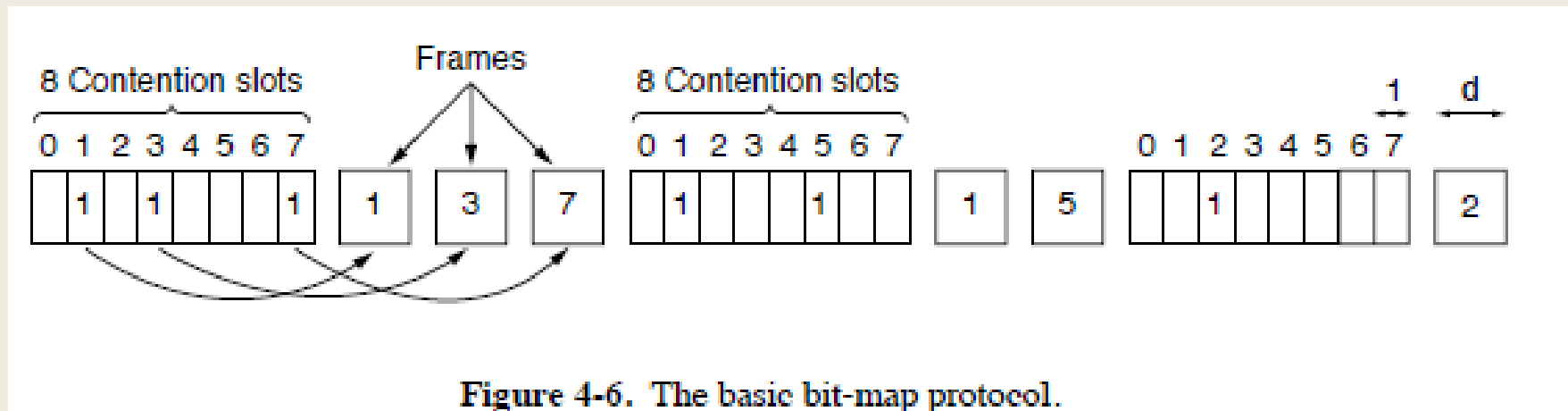
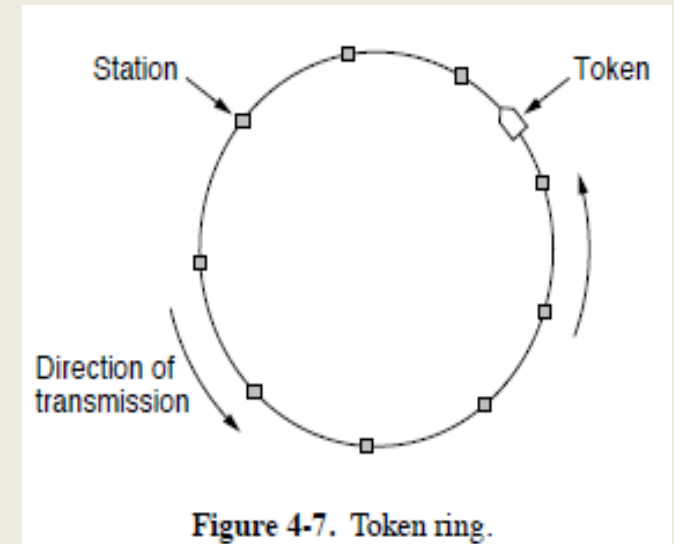


Figure 4-6. The basic bit-map protocol.

Token Passing :

- The essence of the bit-map protocol is that it lets every station transmit a frame in turn in a predefined order.
- Another way to accomplish the same thing is to pass a small message called a token from one station to the next in the same predefined order.
- The token represents permission to send.
- If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station.
- If it has no queued frame, it simply passes the token.
- The stations are connected one to the next in a single ring.
- Passing the token to the next station then simply consists of receiving the token in from one direction and transmitting it out in the other direction.

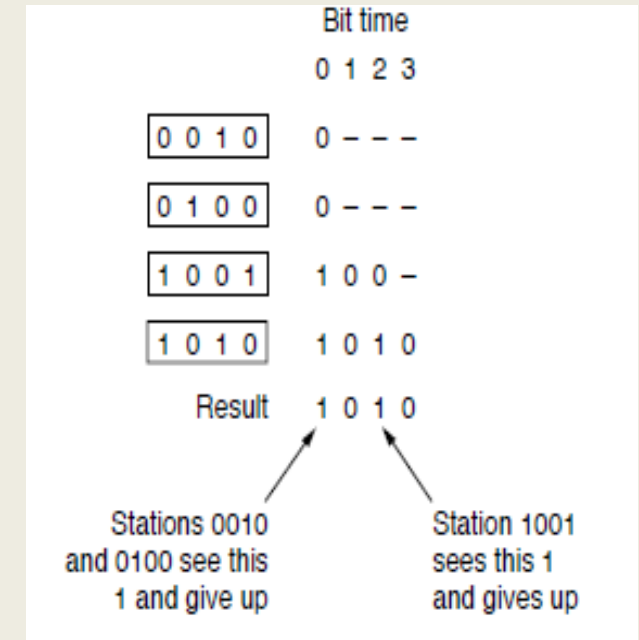


Binary Countdown :

- A problem with the basic bit-map protocol, and by extension token passing, is that the overhead is 1 bit per station, so it does not scale well to networks with hundreds or thousands of stations.
- We can do better than that by using binary station addresses with a channel that combines transmissions in a certain way.
- A station wanting to use the channel now broadcasts its address as a binary bit string, starting with the high-order bit.
- All addresses are assumed to be the same number of bits.
- The bits in each address position from different stations are BOOLEAN ORed together by the channel when they are sent at the same time.

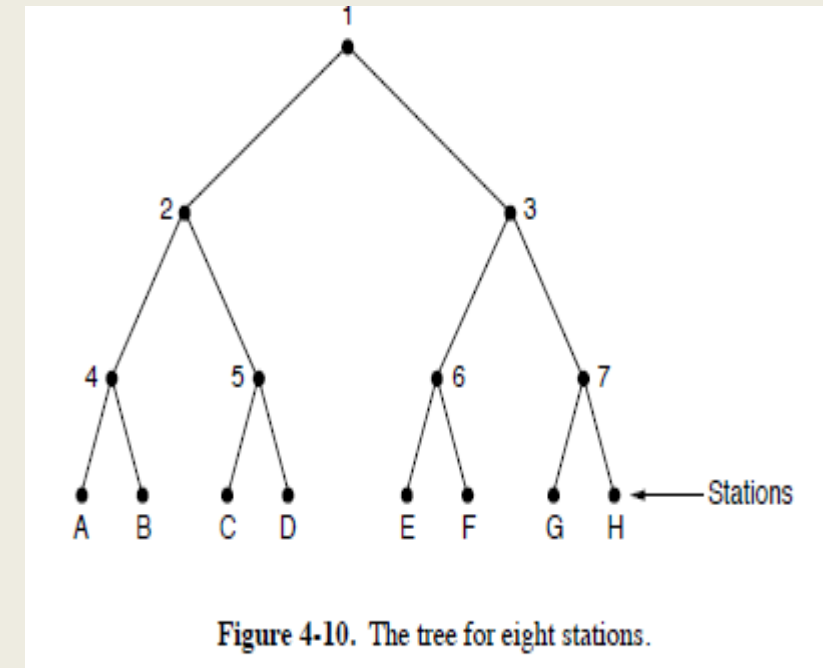
Binary Countdown :

- For example, if stations 0010, 0100, 1001, and 1010 are all trying to get the channel, in the first bit time the stations transmit 0, 0, 1, and 1, respectively.
- These are ORed together to form a 1. Stations 0010 and 0100 see the 1 and know that a higher-numbered station is competing for the channel, so they give up for the current round.
- Stations 1001 and 1010 continue.
- The next bit is 0, and both stations continue.
- The next bit is 1, so station 1001 gives up.
- The winner is station 1010 because it has the highest address



Limited-contention Protocol:

- It would be nice if we could combine the best properties of the contention and collision-free protocols, arriving at a new protocol that used contention at low load to provide low delay, but used a collision-free technique at high load to provide good channel efficiency.
- Such protocols, which we will call limited-contention protocols.
- In the first contention slot following a successful frame transmission, slot 0, all stations are permitted to try to acquire the channel.
- If one of them does so, fine. If there is a collision, then during slot 1 only those stations falling under node 2 in the tree may compete.



IEEE 802:

- Many of the designs for personal, local, and metropolitan area networks have been standardized under the name of IEEE 802.

802.1	LAN/MAN bridging, remote MAC bridging
802.2	Logical Link control standards for connectivity
802.3	Ethernet standards for CSMA/CD
802.4	Standards for token passing bus access
802.5	Standards for token rings & LAN/MAN comm.
802.6	Standards for information exchange
802.7	Broadband LAN cabling
802.8	Fiber Optic connection
802.9	For services like voice and data

ETHERNET:

- Two kinds of Ethernet exist:
- **Classic Ethernet**, which solves the multiple access problem using the techniques we have studied
- **Switched Ethernet**, in which devices called **switches** are used to connect different computers.
- Classic Ethernet is the original form and ran at rates from 3 to 10 Mbps.
- Switched Ethernet is what Ethernet has become and runs at 100, 1000, 10,000, 40,000, or 100,000 Mbps, in forms called fast Ethernet, gigabit Ethernet, 10-gigabit Ethernet, 40-gigabit Ethernet, or 100-gigabit Ethernet.

